

Module 12: Two-Layer Neural Network for Graduate Admissions Prediction

[1] Loading and preprocessing data...

Original rows: 30000
Rows after filtering: 20597
Accepted: 10567
Rejected: 10030
Features: ['gpa', 'gre_quant', 'gre_verbal', 'gre_aw', 'ms_vs_phd', 'international_vs_local']

[2] Splitting data...

Training set size: 16477
Test set size: 4120

[3] Preprocessing features...

Training-set medians:

gpa	3.8
gre_quant	168.0
gre_verbal	159.0
gre_aw	4.0
ms_vs_phd	0.0
international_vs_local	1.0

Training-set means:

gpa	3.900992
gre_quant	171.097105
gre_verbal	158.940280
gre_aw	4.441866
ms_vs_phd	0.444984
international_vs_local	0.696183

Training-set standard deviations:

gpa	0.893102
gre_quant	24.225325
gre_verbal	6.534905

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gre_aw 6.127889
ms_vs_phd 0.496979
international_vs_local 0.459919

[4] Training the neural network...

Epoch 0 | Train MSE: 0.251267 | Test MSE: 0.251039 | Test Acc: 0.4891
Epoch 100 | Train MSE: 0.250051 | Test MSE: 0.249979 | Test Acc: 0.5240
Epoch 200 | Train MSE: 0.249789 | Test MSE: 0.249783 | Test Acc: 0.5090
...
(continue with the full training log from training.log)
...
Epoch 9900 | Train MSE: 0.246892 | Test MSE: 0.247773 | Test Acc: 0.5413

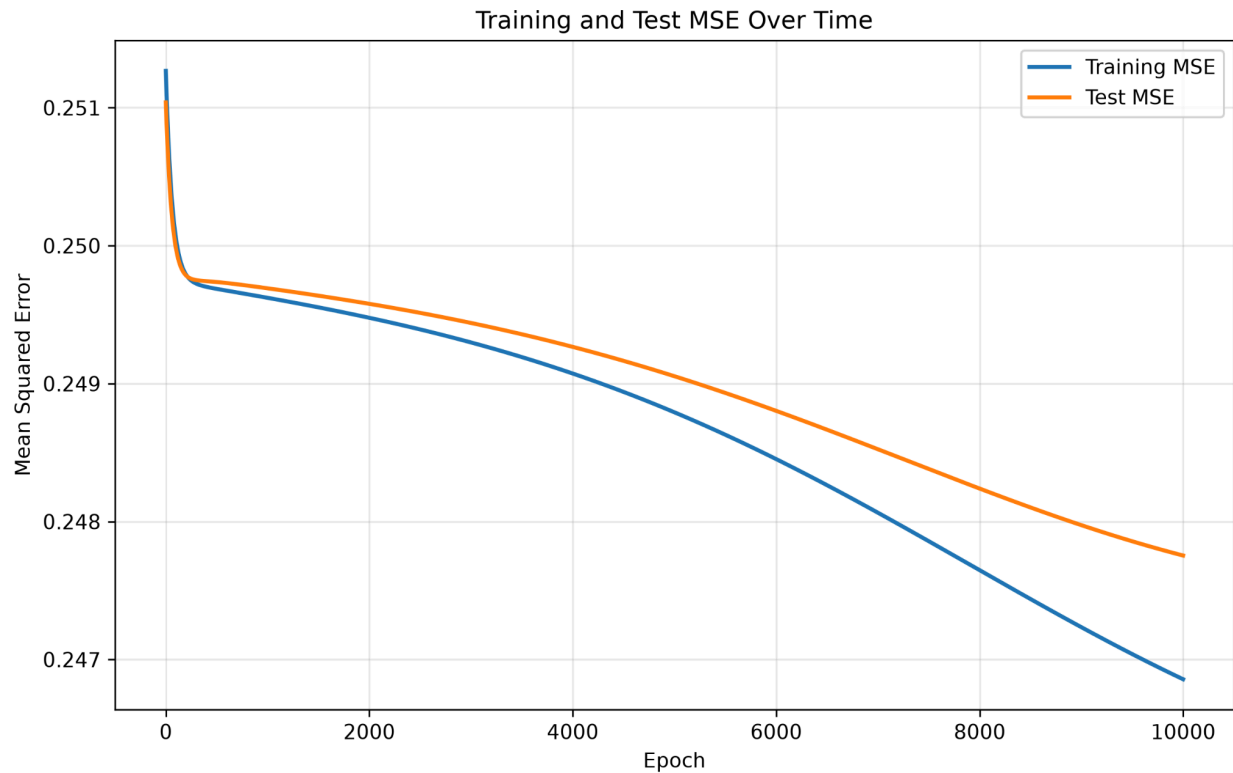
Best test MSE: 0.247754 at epoch 9999

Final Evaluation Results:

Best epoch: 9999
Best test MSE: 0.247754
Final training accuracy: 0.5511 (55.1%)
Final test accuracy: 0.5413 (54.1%)
Rows after filtering: 20597
Train/Test split: 16477 / 4120

Overfitting discussion:

The model does not appear to overfit significantly, as the training and test accuracies are close (55.1% vs 54.1%). However, the overall test accuracy is weak (54.1%), suggesting that the model is barely better than random guessing (50%). This indicates that the dataset and feature set may not be sufficient for a realistic admissions predictor.



Artificial Applicant Predictions:

Applicant 1:

- GPA: 3.9
- GRE Quant: 165
- GRE Verbal: 160
- GRE AW: 5.0
- PhD
- Local
- Predicted Probability: 0.5073
- Predicted Status: Accepted

Applicant 2:

- GPA: 3.2
- GRE Quant: 145
- GRE Verbal: 140
- GRE AW: 3.0
- Masters
- International

- Predicted Probability: 0.5109
- Predicted Status: Accepted

Discussion:

Both applicants were predicted as accepted with probabilities just above 0.5. The model appears to be uncertain in its predictions, as both probabilities are close to the 0.5 threshold. This suggests that the model has not learned strong patterns from the data and may not be reliable for individual predictions.

Reflection:

1. What is useful about implementing a neural network manually?

Implementing a neural network from scratch provides a deep understanding of how forward propagation, backpropagation, and gradient descent work. It demystifies the "black box" of deep learning frameworks and helps build intuition for debugging and improving models.

2. What are the limitations of using MSE for a binary classification task?

MSE is designed for regression tasks and penalizes large errors more heavily. For binary classification, cross-entropy loss is typically preferred because it more directly measures classification performance. MSE can lead to slower convergence and less interpretable probabilities.

3. What information is missing from this admissions dataset?

Important factors such as research experience, letters of recommendation, statement of purpose quality, undergraduate institution prestige, and relevant work experience are not included. These factors often play a significant role in graduate admissions decisions.

4. Why might this model produce misleading results even if test accuracy looks reasonable?

Even if test accuracy is above 50%, the model may be capturing spurious correlations in the data rather than meaningful patterns. The features available (GPA, GRE) are limited and may not fully capture the admissions decision process. Additionally, the dataset may contain biases or imbalances that the model exploits.

5. What would you change to make this model stronger or more realistic?

I would:

- Add more features (research experience, letters of recommendation, statement of purpose quality)
- Use cross-entropy loss instead of MSE
- Add regularization to prevent overfitting
- Use a larger and more representative dataset

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- Try different network architectures and hyperparameters
- Use cross-validation to tune hyperparameters more systematically